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main.py

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Terminal

Terminal

```
1  # WRITE YOUR CODE HERE
2  import os
3  import openai
4  from PIL import Image, ImageOps, ImageChops
5  from io import BytesIO
6  import requests
7
8  # Set environment variables
9  #openai.api_key = os.getenv('OPENAI_KEY')
10 openai.api_key='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'
11
12 # Generate the base image
13 def generate_base_image(prompt):
14     response = openai.Image.create(
15         prompt=prompt,
16         n=1,
17         size="512x512"
18     )
19     return response['data'][0]['url']
20
21 def download_image(image_url,x):
22     response = requests.get(image_url)
23     img_data = response.content
24     img = Image.open(BytesIO(img_data))
25     with open(x+'.jpg', 'wb') as handler:
26         handler.write(img_data)
27     return img
28
29 def generate_earth_image(angle):
30     prompt = f"Realistic Earth from space at {angle} degrees angle"
31     image_url = generate_base_image(prompt)
32     image_filename = f"earth_{angle}_degrees"
33     image = download_image(image_url, image_filename)
34     return image
35
36 angles = range(0, 360, 10)
37 earth_images = [generate_earth_image(angle) for angle in angles]
```

Guide

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Collapse

<>⚙️☰

Intro

To generate a sequence of images, we can loop through different prompts or modify the prompt with changing properties. In this case, we'll generate images of Earth from different angles. Here's an example code snippet to generate the sequence of images:

```
def generate_earth_image(angle):
    prompt = f"Realistic Earth from space at {angle} degrees angle"
    image_url = generate_base_image(prompt)
    image_filename = f"earth_{angle}_degrees"
    image = download_image(image_url, image_filename)
    return image

angles = range(0, 360, 10)
earth_images = [generate_earth_image(angle) for angle in angles]
```

In this code, we define a function `generate_earth_image` that takes an `angle` as an argument and generates an image of Earth from that angle using the DALL-E 2 API. We then create a list of angles from 0 to 350 degrees in increments of 10 degrees and generate a corresponding image for each angle.

The files are provided for you because the code might take a bit to load since generating and saving a ton of pictures. IF YOU WANT YOU CAN REGENERATE THEM USING THE TRY IT BUTTON BELOW. Not recommended

Here is a sample of the earth at 230 degrees.



TRY IT

After Generating your images please comment out the code that would generate the images.

"Create a list of angles from 0 to 350 degrees in increments of 10 degrees and generate a corresponding image for each angle". How many angles are we generating ?

☒ 36

☐ 350

☐ 5

☐ 26

We will be generating 36 angles in total. Here is the list of angles:

0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 310, 320, 330, 340, 350

Check It!

Next

100% (37:65)

https://codio.com/sandipan-dey/animating-generated-images/tree/main.py